



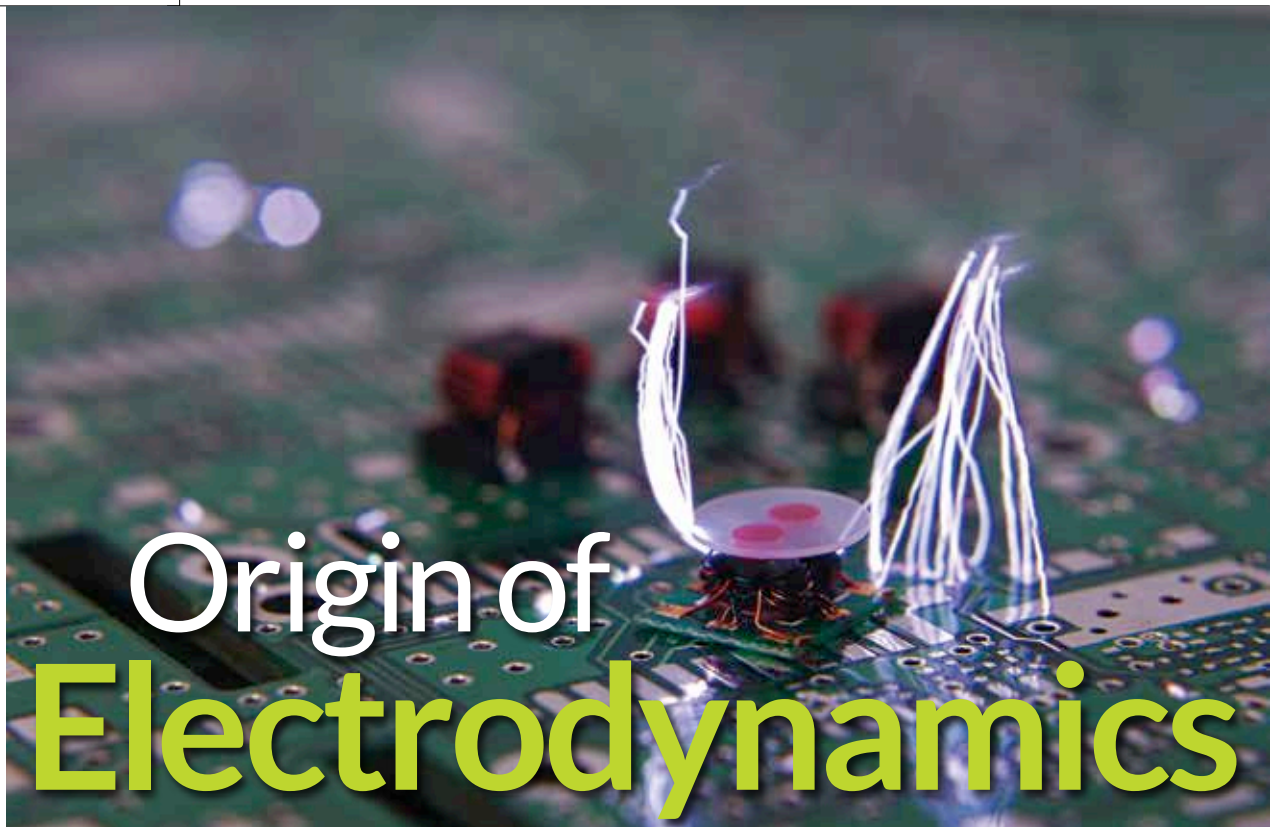
NASA's Perseverance Rover

The rover has landed on Mars' Jerezo Crater and shows a panoramic view with immense detail.
<http://digit.in/mar21-91>



Blue Origin

Jeff Bezos' giant rocket is all set to launch late in 2022.
<http://digit.in/mar21-92>



The history of electrodynamics and the foundation of its theory

Kanishk Rathod | feedback@digit.in

The theoretic formulation and the absolute clear conceptual ordering of the electromagnetic phenomena was with the theory of James Clerk Maxwell (in 1864). This was by far the greatest single leap in the human knowledge of electricity. He presented a simpler form of Faraday's previous work and condensed all the current knowledge into a linked set of differential equations. From this he was also able to calculate the possibility of electromagnetic waves. His theory involved the theory of light, in the form of a velocity constant, and thus proved electrophysics and optics were connected, in fact, optics became a niche

subfield of electrophysics. This was an extraordinary result for theoretical physics, however it wasn't well received in Europe. The reason was that this theory was based on Faraday's work who was unfamiliar to mechanically trained thoughts. Many prominent scientists at the time (such as L. Boltzmann) tried to explain the correlations through mechanic models, however, the field theory proved victorious over all others. This was finally confirmed in 1888, when Heinrich Hertz experimentally discovered electrically produced waves, thus proving that light waves have the same characteristics as electric waves.

By now the third law of thermodynamics by Nernst was widely known, which stated that absolute zero was unattainable. Research in electrical phenomena predicted that resistance of electrical currents would drop in extremely low temperatures, which was documented by Olszewski and Wroblwesi in the 1800s which was followed by the milestone of achieving superconductivity through liquified helium by Onnes in 1908. Between 1900-10 most prominent

scientists such as Wilhelm Wien, Max Abraham, Hermann Minkowski etc believed that all forces of nature were electromagnetic in essence. Hendrik Lorentz's work on the electron theory connected this, which introduced the separation of matter at the sub-atomic levels. The field theory further progressed and velocity changes in this field were capped at the speed of light.

Henri Poincaré went on to continue the work of Hendrik Lorentz and formulated the principle of relativity to incorporate it with electrodynamics – more specifically, he used the constant speed of light to become a convention for simplifying the laws of nature. In 1905, he went on to declare the principle of relativity a general law of nature. He also suggested that non-electrical forces are in play to stabilize the electron within an atom, and insisted on gravitation being a separate non-electrical force which was contrary to the view held by the believers of the electromagnetic theory. However, he still used the notion of an ether and understood the difference between apparent and real time, and hence did not